



Transplantation of 3D Printed Dorsal Spinal Neural Progenitor Cell Scaffolds for Spinal Cord Injury

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BACKGROUND

Regional Identity in the Spinal Cord

- Spinal cord injury (SCI) is associated with **sensory** and **motor** deficits that arise due to damage in the **dorsal** and **ventral** spinal cord, respectively.
- Dorsal horn sensory circuitry remains understudied despite the majority of SCI patients experiencing sensory dysfunction.

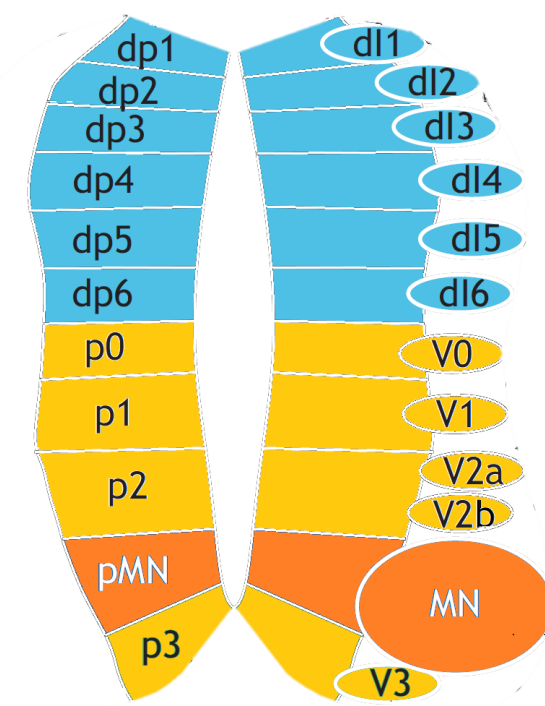
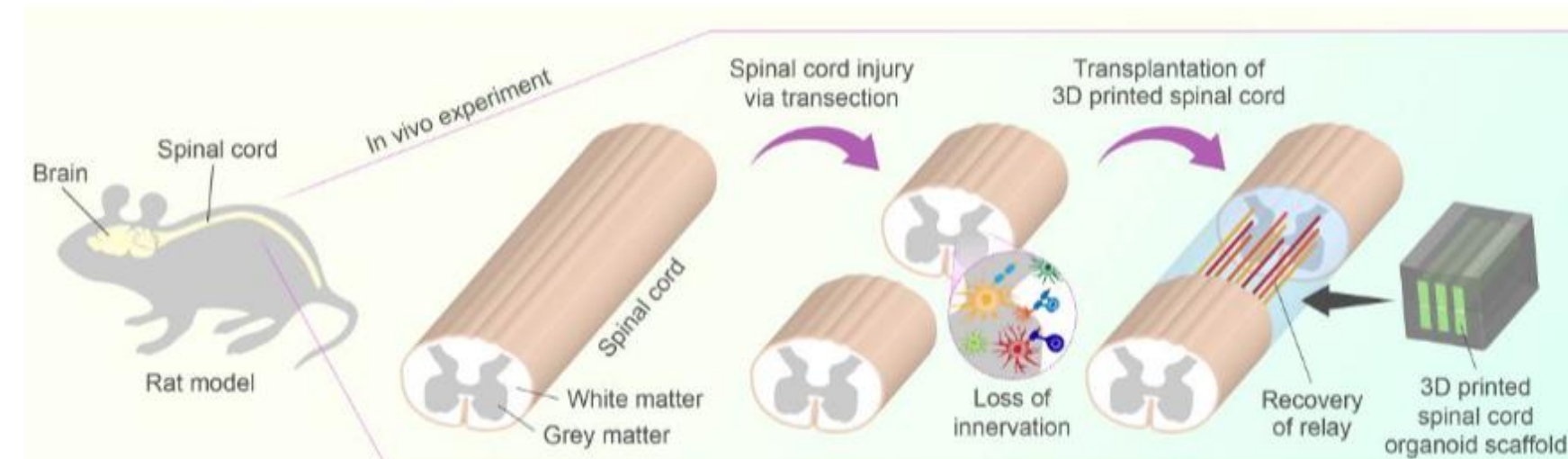


Figure 1. Regional progenitor domains in the developing spinal cord (left) and the mature cell populations they give rise to (right). In SCI, damage to the dorsal cord (top/blue) is most often associated with sensory impairment, while damage to the ventral cord (bottom/orange) is associated with motor impairment.

Promoting Integration after Cell Transplantation

- Cell transplantation utilizes stem cell derived neural progenitor cells to restore lost connections and promote sensorimotor recovery following SCI, though poor graft integration limits functional improvement.
 - Regional specification:** Patterning of grafted cells to the phenotypic identity at the level of the injury has been shown to improve integration¹.
 - Structural support:** The use of scaffolding to provide support and directed growth can improve integration after transplantation.

Figure 2. Rodent cell transplantation model incorporating regionally specific cell grafts and spinal scaffolding. In our lab, we utilize an athymic nude rat (ATN) model with a complete transection SCI.



METHODS

Establishment of Dorsal Spinal Neural Progenitor Cells (dsNPCs)

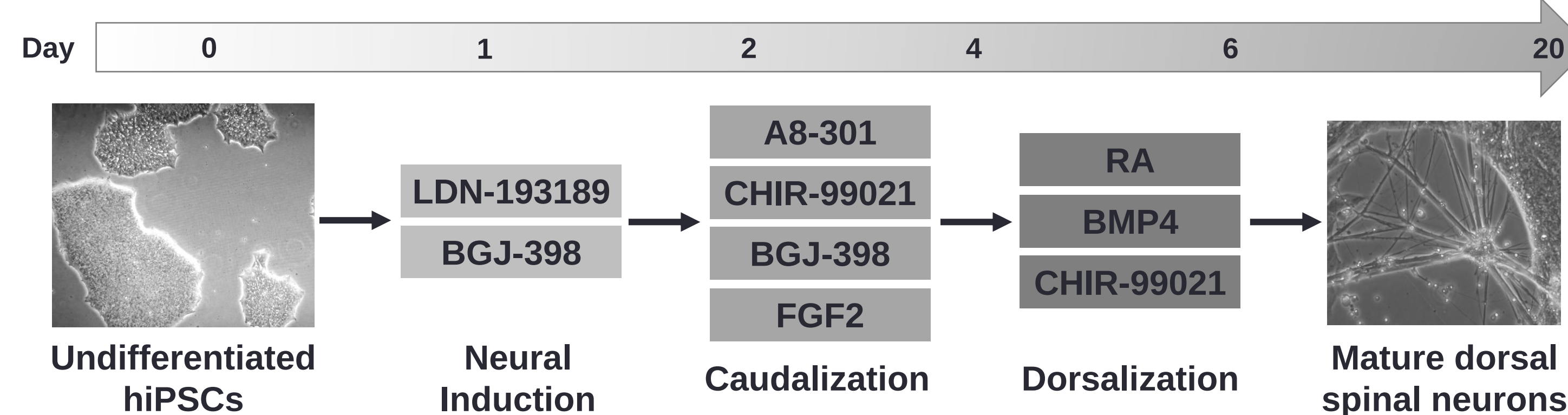


Figure 3. Directed differentiation of dorsal spinal neurons^{2,3}. Following an accelerated 24-hour neural induction, cells may be dorsalized or ventralized (data not shown) to neural progenitors at day 6 or post-mitotic at day 20.

3D Printing of dsNPC Scaffolds

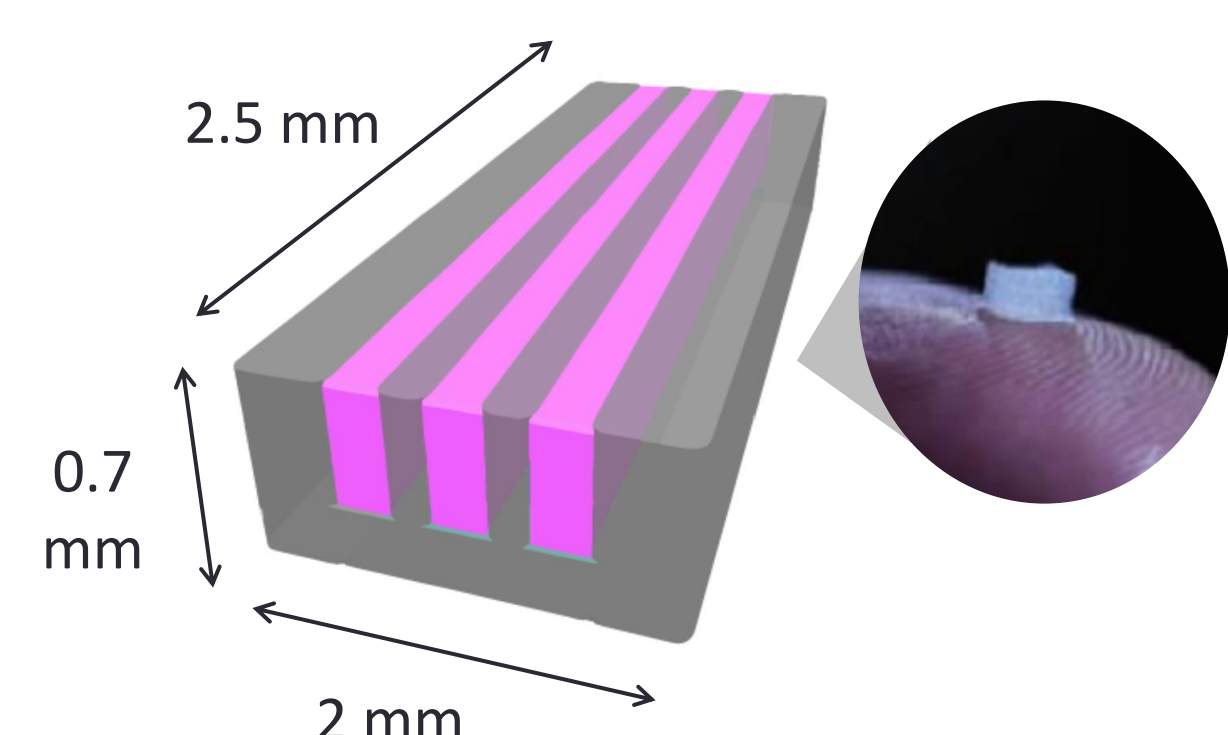


Figure 4. Construction of silicone scaffolds and sNPC bioink⁴. Matrigel bioink (1:1) is printed at a density of 2×10^8 cells/mL into three 200µm scaffold channels.

Acute Transection SCI Model

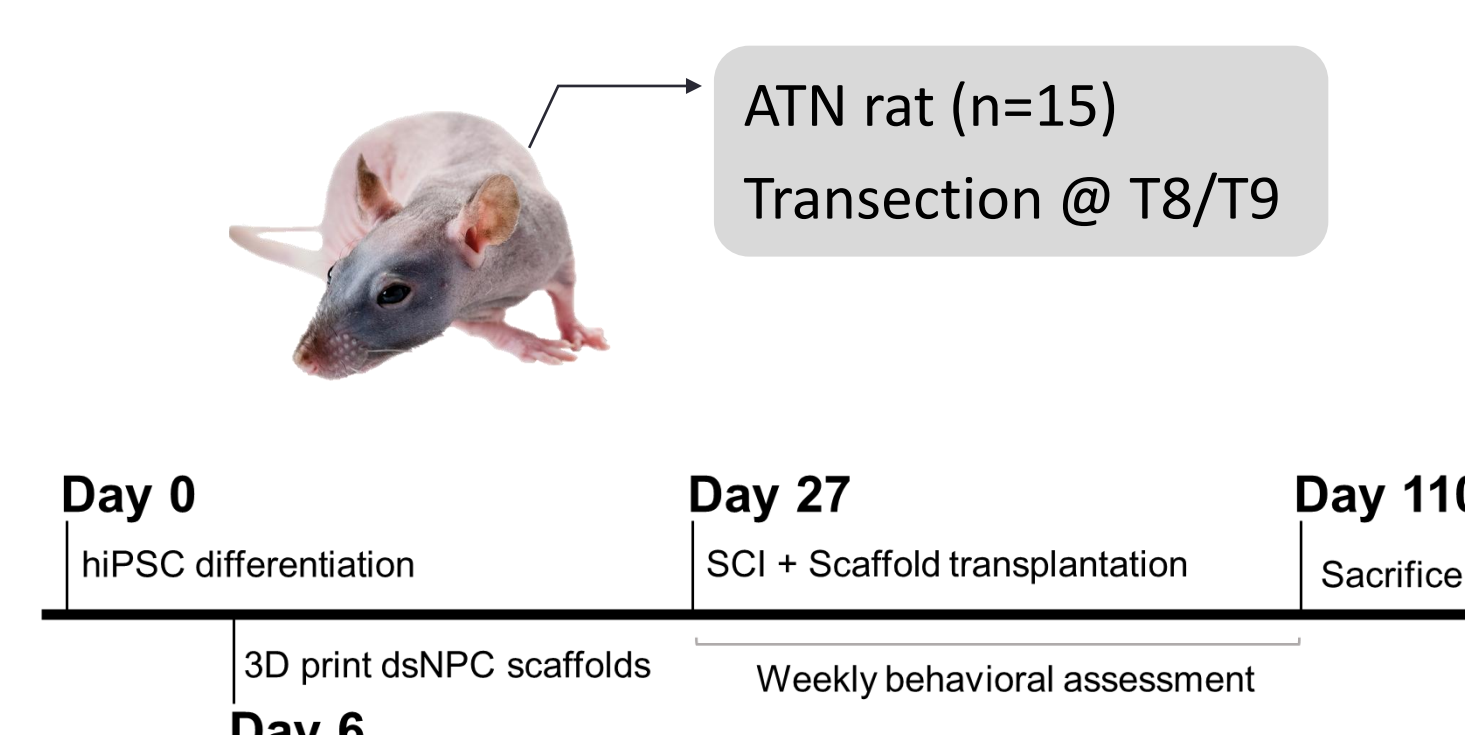
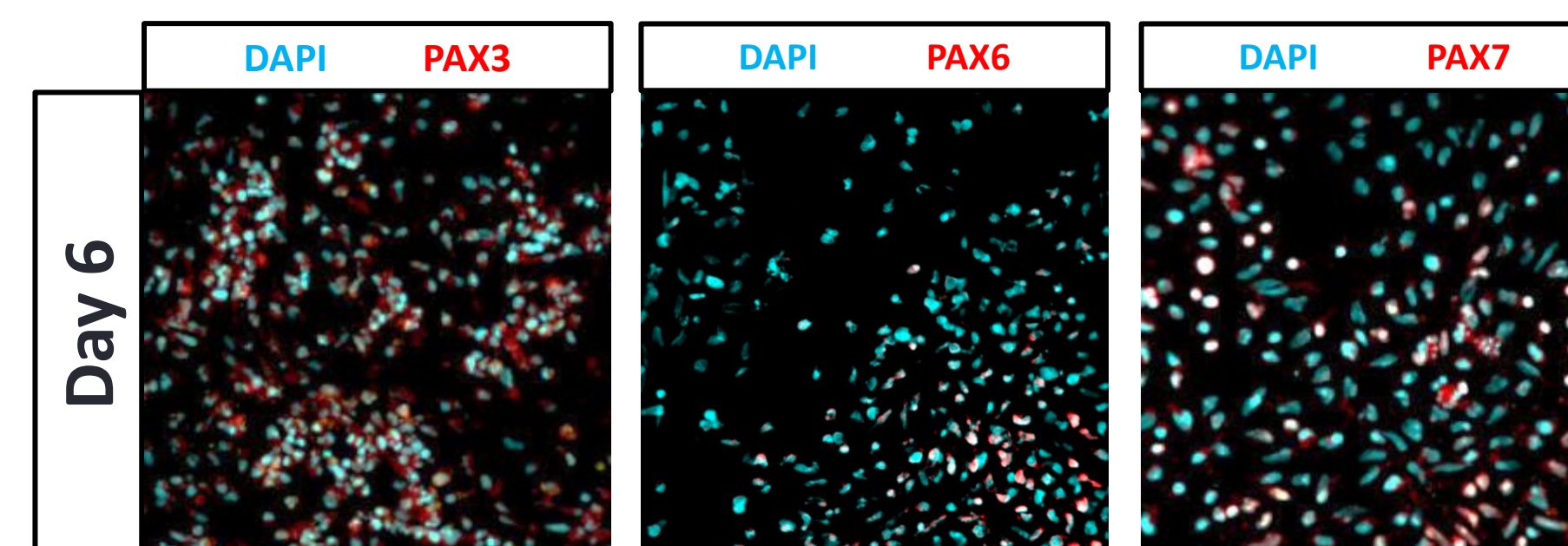


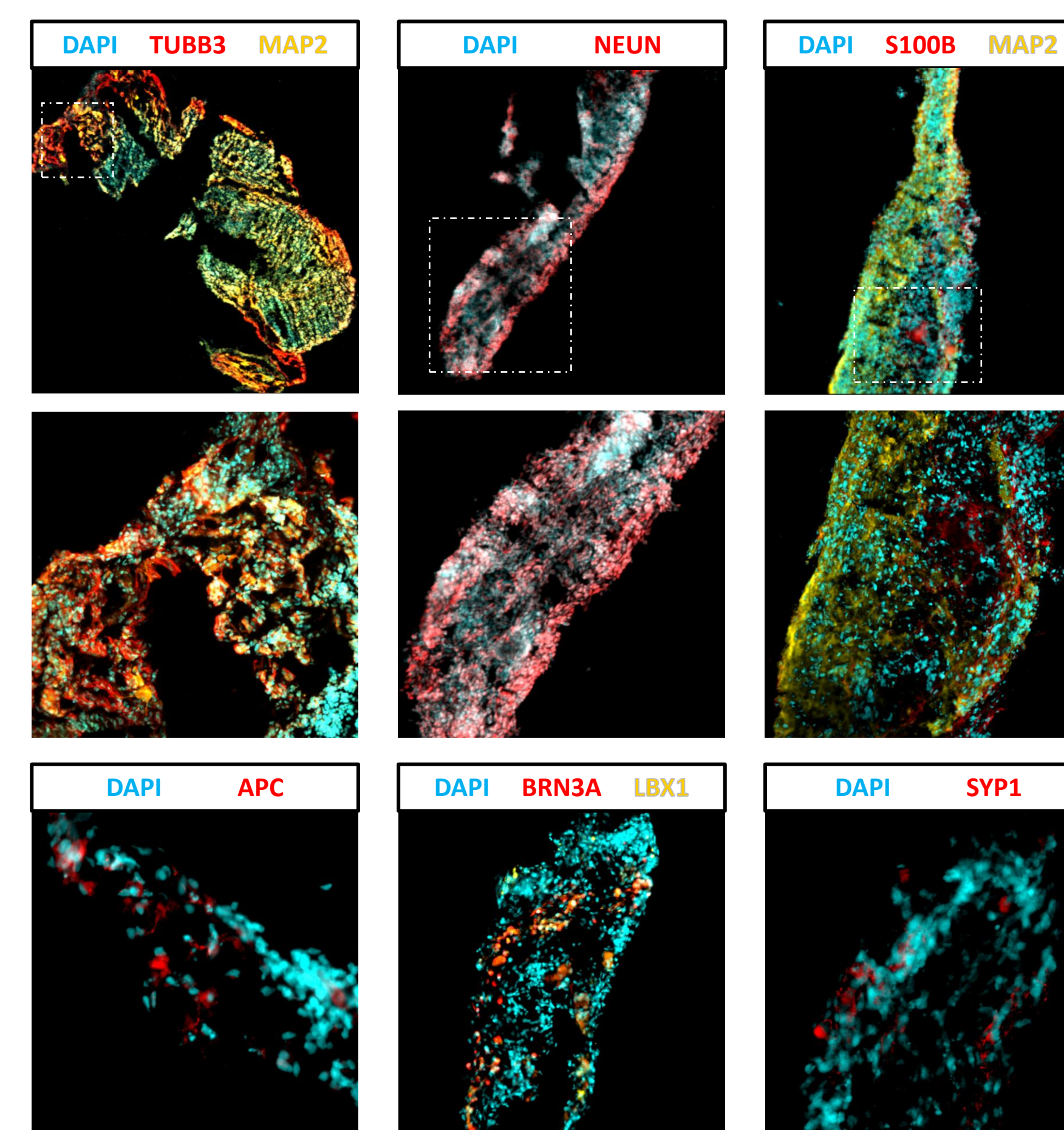
Figure 5. Timeline of rodent study. ATN rats received a complete transection SCI followed by immediate scaffold transplantation. Behavior was assessed for 12 weeks post-injury.

Dorsal sNPCs undergo 6 days of directed differentiation prior to printing into scaffolds



At Day 6, dsNPCs exhibit PAX gene expression consistent with a dorsal progenitor phenotype.

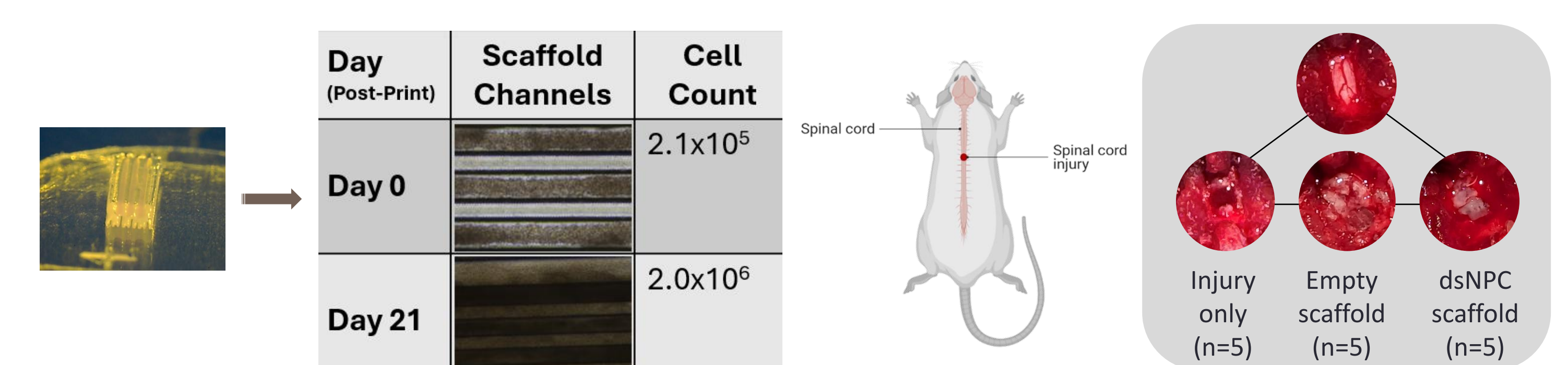
Mature scaffolds express markers consistent with neuron, astrocyte, and oligodendrocyte identity



At Day 21, cells are mostly neuronal (60-80%) with mature dorsal and presynaptic markers.

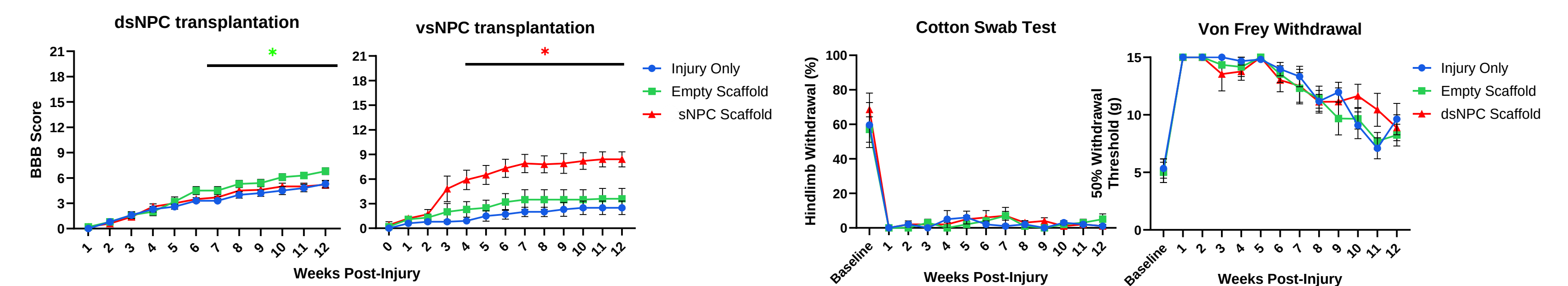
RESULTS

Cell-laden scaffolds were matured for 21 days after printing and transplanted into a complete transection SCI at T8/T9 in fifteen female athymic nude (ATN) rats



Cells filled the channels and formed complex 3D assembloids. Two scaffolds were sandwiched together to form a single implant for transplantation. A 2.5mm segment of spinal cord was removed followed by immediate insertion of the scaffold implant.

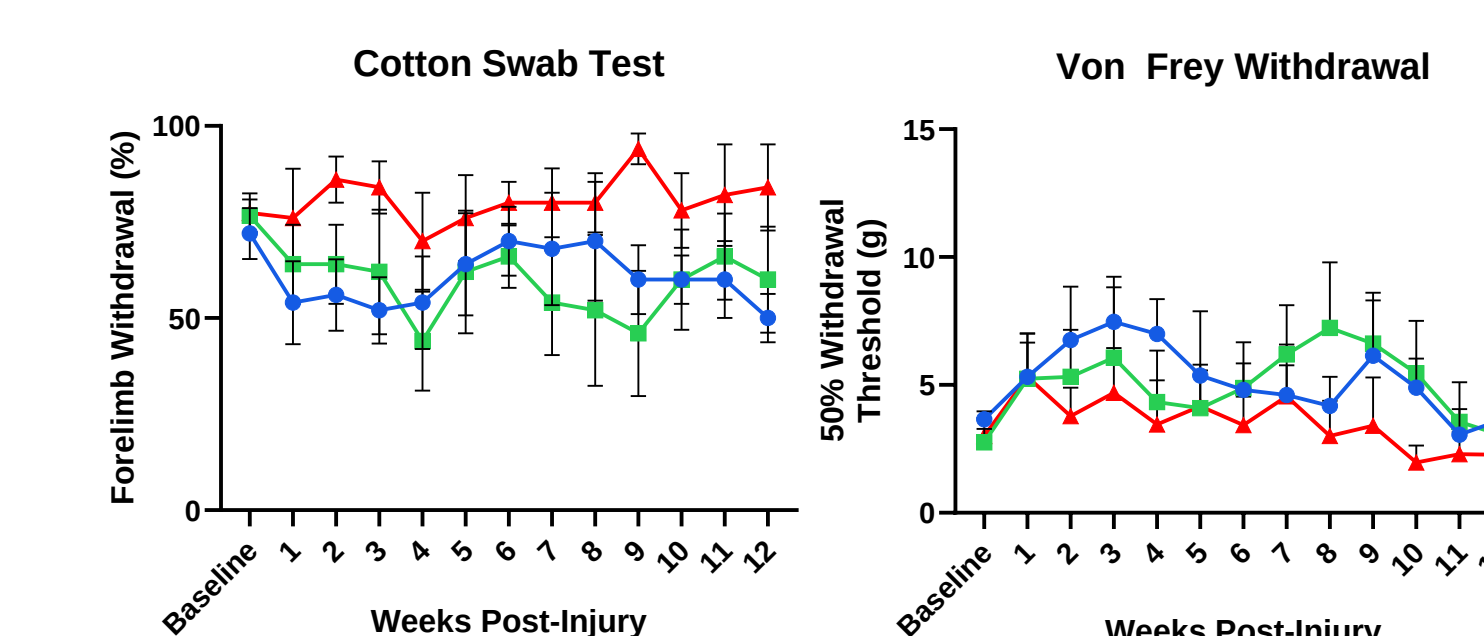
Sensorimotor function of the hindlimbs was evaluated weekly using open field locomotor scoring and mechanical sensitivity testing for 12 weeks post-SCI



Regional specificity of cells influences graft integration and functional recovery. Ventral but not dorsal transplants promote hindlimb motor recovery (Two-Way ANOVA, $p < 0.05$).

Graft-mediated recovery of hindlimb sensory function is not detectable 12 weeks post-SCI. No significant differences between treatment conditions observed.

Altered above-level sensitivity in the forelimb with dsNPC scaffolds



Trend of decreased sensitivity after injury in left forelimb was attenuated in animals with dsNPC scaffolds.

FUTURE DIRECTIONS

- Our lab utilizes a combinatorial approach with regionally specified transplants and 3D printed bio-scaffolds to improve integration in a rodent model of SCI.
- We have successfully 3D-printed dorsal and ventral sNPCs into a bioscaffold for implantation in acute and chronic transection SCI rat models and observed functional and anatomic recovery following ventral transplantation. We are now examining the histology for the dsNPC transplantation to evaluate the differentiation and integration of cells and study possible sources of above-level behavioral effects, such as neuroinflammation and axonal dieback.
- Future research will utilize both dorsal and ventral sNPC populations in tandem to evaluate effects on functional sensorimotor recovery.

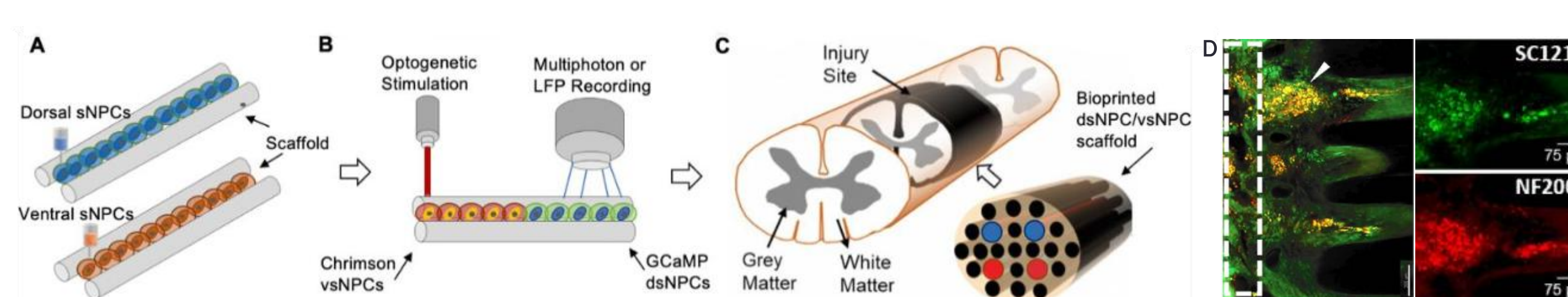


Figure 6. Future development and functional characterization of regionally specific spinal cord bioscaffolds for use in SCI. A. Dorsal and ventral sNPCs are optimized for 3D-printing into scaffolds. B. Optogenetic and calcium imaging techniques will be utilized to study the conditions for optimal neural network formation. C. Dorsal and ventral populations will be transplanted in tandem in an organotypic arrangement. D. Integration of sNPC scaffolds with host networks *in vivo*.

References

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